A FINITE LOCALLY FREE MODULE TRIVIAL AT ALL CODIMENSION ONE POINTS OF A NORMAL BASE IS PULLED BACK FROM THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EXG (sources/stacks-flat/pinned-original-source.tex line 12641).

Allowed standard prerequisites (editorial).

- Flat base change for cohomology. Cohomology of Schemes, Lemma labelled coherent-lemma-flat-base-change-cohomology, cited at proof lines 12657-12659. Transfers the isomorphism of the canonical map to the localisations at codimension one points.
- Checking an isomorphism by depth and associated points. Divisors, Lemma labelled divisors-lemma-check-isomorphism-via-depth-and-ass, cited at proof lines 12663-12665. Reduces the isomorphism to a depth bound at points of codimension at least two.
- Normality, depth and pushforward. Properties, Lemma labelled properties-lemma-criterion-normal, cited at proof lines 12673-12675; Divisors, Lemma labelled divisors-lemma-depth-pushforward, cited at proof lines 12675-12677; Algebra, Lemma labelled algebra-lemma-apply-grothendieck, cited at proof lines 12669-12671. Supplies depth(O_{S,s}) at least 2 and the induced bound for f_*E.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12661 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AVM: sources/stacks-chow/tags.txt, source lines 13690–13690. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12665 of the counted proof. It is

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Source reference. Original lemma 0337: sources/stacks-chow/tags.txt, source lines 4046–4046. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12672 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0345: sources/stacks-chow/tags.txt, source lines 4080–4080. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12676 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EY0: sources/stacks-chow/tags.txt, source lines 18675–18675. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12678 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module. Assume*

(1) *f is flat and proper and $\mathcal{O}_S = f_*\mathcal{O}_X$,*
(2) *S is a normal Noetherian scheme,*
(3) *the pullback of $\mathcal{E}$ to $X \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ is free for every codimension 1 point $s \in S$.*

Then $\mathcal{E}$ is isomorphic to the pullback of a finite locally free $\mathcal{O}_S$-module.

Proof. We will prove the canonical map

$$\Phi : f^*f_*\mathcal{E} \longrightarrow \mathcal{E}$$

is an isomorphism. By flat base change (Cohomology of Schemes, Lemma 02KH) and assumptions (1) and (3) we see that the pullback of this to $X \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ is an isomorphism for every codimension 1 point $s \in S$. By Divisors, Lemma 0AVM it suffices to prove that $\mathrm{depth}((f^*f_*\mathcal{E})_x) \geq 2$ for any point $x \in X$ mapping to a point $s \in S$ of codimension ≥ 2. Since f is flat and $(f^*f_*\mathcal{E})_x = (f_*\mathcal{E})_s \otimes_{\mathcal{O}_{S,s}} \mathcal{O}_{X,x}$, it suffices to prove that $\mathrm{depth}((f_*\mathcal{E})_s) \geq 2$, see Algebra, Lemma 0337. Since S is a normal Noetherian scheme and $\dim(\mathcal{O}_{S,s}) \geq 2$ we have $\mathrm{depth}(\mathcal{O}_{S,s}) \geq 2$, see Properties, Lemma 0345. Thus we get what we want from Divisors, Lemma 0EY0. □